

PE-EC702C — Neural Network & Fuzzy Logic Control

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Module 5 Exam Notes: Applications of Neural Networks & Fuzzy Logic

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Applications of Artificial Neural Networks

Neural networks excel at pattern recognition, function approximation, and adaptive modeling due to their parallel distributed architecture and learning capabilities.

1. **Pattern and Character Recognition:** ANNs (particularly CNNs and MLPs) are widely used for Optical Character Recognition (OCR), signature verification, and face recognition by learning translation-invariant spatial features.
2. **Image Compression:** Feedforward networks with a narrow hidden layer (bottleneck autoencoders) perform dimensionality reduction. The input image is reconstructed at the output layer, while the compressed representation is stored at the hidden layer.
3. **Communication Systems (Channel Equalization):** Equalizers modeled using neural networks (like radial basis function networks) reduce Inter-Symbol Interference (ISI) in dispersive channels by acting as non-linear classifiers.
4. **Control Systems (System Identification):** ANNs model the forward and inverse dynamics of complex, non-linear chemical and mechanical processes, facilitating adaptive and predictive control schemes.

2. Applications of Fuzzy Logic

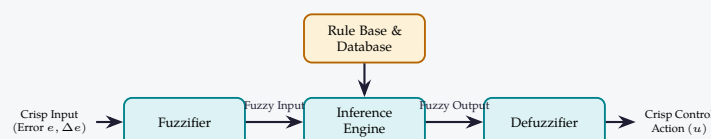
Fuzzy logic is highly effective in systems involving linguistic variables and human decision-making processes, where exact mathematical models are unavailable.

- **Fuzzy Pattern Recognition:** Classifies inputs into overlapping classes based on membership values rather than crisp thresholds.
- **Fuzzy Image Compression:** Uses fuzzy rules to adaptively select quantization levels across different image blocks based on local edge/texture complexity.

3. Fuzzy Logic Controllers (FLC)

A Fuzzy Logic Controller (FLC) implements a control algorithm by evaluating heuristic, rule-based statements rather than mathematical differential equations.

Fuzzy Logic Controller Block Diagram



3.1 Standard FLC Architecture Components

1. **Fuzzifier (Fuzzification Interface):** Measures physical inputs and converts them into fuzzy values (linguistic terms like *Positive Big*, *Zero*, *Negative Small*) using

predefined input membership functions.

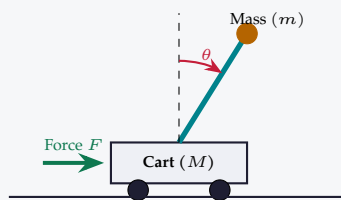
2. **Knowledge Base (Rule & Database):** Contains the database (definitions of membership functions) and the Rule Base (a collection of linguistic control statements in **IF-THEN** format).
3. **Inference Engine (Decision Making Logic):** Evaluates the active rules to determine the overall fuzzy control output based on composition rules (usually Max-Min composition).
4. **Defuzzifier (Defuzzification Interface):** Converts the combined fuzzy output into a single crisp control command to drive physical actuators.

4. Control System Case Studies

4.1 Inverted Pendulum Controller

The goal is to maintain a rotating pole in an upright position by moving a motorized cart horizontally.

Inverted Pendulum Physical Model



- **Inputs:** Angle error θ (deviation from vertical) and angular velocity $e_{\omega} = d\theta/dt$.
- **Output:** Motor force F applied to the cart.
- **Rule Base example:**
 - IF angle (θ) is Positive Small AND angular velocity ($d\theta/dt$) is Zero, THEN force (F) is Positive Small.

4.2 Fuzzy Washing Machine Controller

An FLC optimizes the wash cycle time, water temperature, and spin speed based on:

- **Inputs:** Dirtiness level of clothes (using optical sensors that measure water transparency) and weight of load.
- **Output:** Cycle duration (wash time).
- **Rule Base example:**
 - IF dirtiness is High AND weight is Medium, THEN wash time is Long.

5. Worked Examples

Fuzzy Inference System Evaluation

Problem: Consider a simple Fuzzy Logic Controller with two input variables x (error) and y (change of error), and one output z (control force). The fuzzy rule base contains the following two rules:

- **Rule 1:** IF x is A_1 AND y is B_1 , THEN z is C_1
- **Rule 2:** IF x is A_2 AND y is B_2 , THEN z is C_2

Given crisp input values $x_0 = 4$ and $y_0 = 6$:

- Membership values: $\mu_{A_1}(x_0) = 0.6$, $\mu_{A_2}(x_0) = 0.3$, $\mu_{B_1}(y_0) = 0.8$, $\mu_{B_2}(y_0) = 0.4$
- Output membership centers (represented by singletons): $z_{C_1} = 20$, $z_{C_2} = 50$

Evaluate the output control value z^* using the **weighted average** defuzzification method.

Solution:

1. Compute Firing Strengths (α -levels) of the rules: Since the connective is **AND**, we use the Min operator:

- For **Rule 1**:

$$\alpha_1 = \min(\mu_{A_1}(x_0), \mu_{B_1}(y_0)) = \min(0.6, 0.8) = 0.6$$

- For **Rule 2**:

$$\alpha_2 = \min(\mu_{A_2}(x_0), \mu_{B_2}(y_0)) = \min(0.3, 0.4) = 0.3$$

2. Calculate Defuzzified Output (z^*): Using the weighted average formula:

$$\begin{aligned} z^* &= \frac{\alpha_1 z_{C_1} + \alpha_2 z_{C_2}}{\alpha_1 + \alpha_2} \\ &= \frac{0.6(20) + 0.3(50)}{0.6 + 0.3} \\ &= \frac{12 + 15}{0.9} = \frac{27}{0.9} = 30 \end{aligned}$$

The output control force is $z^* = 30$.

Quick Revision — Module V

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Key Formulas at a Glance

Rule Firing Strength (AND) $\alpha_i = \min(\mu_{A_i}(x_0), \mu_{B_i}(y_0))$

Rule Firing Strength (OR) $\alpha_i = \max(\mu_{A_i}(x_0), \mu_{B_i}(y_0))$

Weighted Average Defuzzification $z^* = \frac{\sum \alpha_i z_i}{\sum \alpha_i}$

Inverted Pendulum Angle Error $e(t) = \theta_{ref} - \theta(t)$

Inverted Pendulum Error Change $\Delta e(t) = e(t) - e(t-1)$

One-Line Definitions

- **Fuzzy Controller:** A control device that uses linguistic rules and fuzzy reasoning to calculate system control inputs.
- **Fuzzifier:** The interface in an FLC that maps physical crisp sensor readings to linguistic fuzzy values.
- **Inference Engine:** The decision-making unit that evaluates active rule statements to determine control signals.
- **Linguistic Variable:** A variable whose values are words or sentences in a language rather than numbers (e.g. "Warm").
- **System Identification:** The process of developing a mathematical or neural network model to describe dynamic process behavior.

Frequently Asked / Viva Questions

1. **What are the main advantages of a Fuzzy Logic Controller over a PID controller?**
 FLCs do not require an exact mathematical model of the process, can handle non-linear dynamics easily, and can incorporate human operator heuristics directly using linguistic rules.
2. **State the function of the FLC Rule Base.**
 The Rule Base contains the control logic written as a set of IF-THEN rules that describe the relationship between input error conditions and output control actions.
3. **How is a neural network used for image compression?**
 An autoencoder network is trained to replicate inputs at its output layer. A bottleneck hidden layer (fewer nodes) forces the network to learn a compressed representation of the image.
4. **What is channel equalization in communications, and how do ANNs help?**

Frequently Asked / Viva Questions (contd.)

Channel equalization compensates for signal distortion (ISI) caused by transmission channels. ANNs act as adaptive classifiers that identify non-linear decision boundaries to recover transmitted symbols.

5. Explain the wash time optimization in a fuzzy washing machine.

Sensors measure the quantity of load and the dirtiness of the water (transparency). The FLC infers the optimum wash cycle time: muddy water with a heavy load triggers a long wash cycle.

6. What inputs are required to balance an inverted pendulum?

The deviation angle θ from the vertical axis (error) and the rate of change of the angle $d\theta/dt$ (angular velocity).

7. What is the difference between Mamdani and Sugeno fuzzy inference systems?

Mamdani systems use fuzzy sets for output membership functions, requiring defuzzification. Sugeno systems use linear mathematical functions or constants for outputs, simplifying calculation.

8. Define a Singleton membership function.

A singleton is a membership function that has a value of 1.0 at a single specific point and 0 everywhere else.

9. How does a neural network model non-linear control systems?

By acting as a system identification model, learning the mapping between system input history and corresponding output responses.

10. What is the role of the database in a fuzzy controller?

The database holds the exact parameters (shapes, coordinates, limits) of the membership functions defined for the input and output variables.

Comparison of Control Strategies

Parameter	Classical PID Control	Fuzzy Logic Control
Process Model	Requires exact mathematical model	No mathematical model required
Control Logic	Defined by differential equations	Defined by linguistic IF-THEN rules
Tuning Parameters	Proportional, Integral, Derivative gains	Membership functions and rule parameters
Non-linear Handling	Difficult (often requires linearization)	Inherently handles non-linearities